Kinematics 1

$V_{avg} = \frac{\Delta x}{\Delta t}$ $a_{avg} = \frac{\Delta v}{\Delta t}$

$V_{inst} = \frac{dx}{dt}$ $a_{inst} = \frac{dv}{dt}$

Core

- $\Delta X = V_0 t + \frac{1}{2} a t^2$
- $V_f^2 = V_0^2 + 2a\Delta X$
- $V_f = V_0 + at$

Vector Subtraction invert

$\vec{A}$ or $\vec{B}$, $\vec{A} = -\vec{A}$

Unit Vectors

$\vec{A} + \vec{B} = (A_x + B_x)\hat{i} + (A_y + B_y)\hat{j}$

$\vec{A} - \vec{B} = (A_x - B_x)\hat{i} + (A_y - B_y)\hat{j}$

2 Forces

$1N = 1kg \cdot \frac{m}{s^2}$

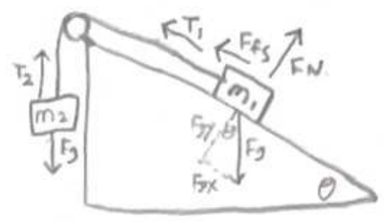
Newton:

① $\Sigma F = 0$, then $\vec{a} = 0$

② $\Sigma F = ma$

③ $F_{AB} = -F_{BA}$

Free Body Diagrams:



$F_{fs} = \mu_s F_N$ *Point of slipping when $F_A = F_{fs}$

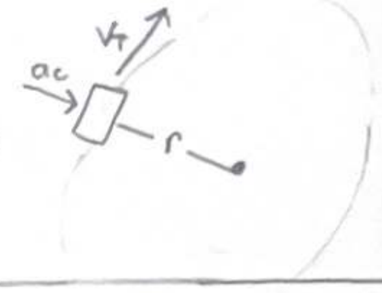
$F_{fk} = \mu_k F_N$ *$\mu_k < \mu_s$

Centripital: $a_c = \omega^2 r$

$F_c = \frac{mv^2}{r}$ $a_c = \frac{v^2}{r}$

$F_c = \frac{m4\pi^2 r}{T^2}$ $V_T = \omega r$

$a_T = r\alpha$



$a_{TOT} = \sqrt{a_c^2 + a_T^2} = \sqrt{r^2\omega^4 + r^2\alpha^2}$

3 Work / Energy

$U_g = -\frac{Gm_1m_2}{r}$

$W_{nc} = \Delta KE + \Delta PE$

$E_i = E_f + \Delta h$ *losses

$W = F\Delta X$

Core:

$E_p = mgh$ $E_k = \frac{1}{2}mv^2$

$E_{ps} = \frac{1}{2}kx^2$

$F_{spring} = -kx$

$Power = \frac{W}{\Delta t}$

$= F \cdot V_{avg}$

watt = J/s

5

4 Momentum

Centre of mass

$x_{cm} = \frac{\Sigma m_i x_i}{m_{total}}$

$y_{cm} = \frac{\Sigma m_i y_i}{m_{total}}$

- $P = mv$

Impulse

- ΔP
- $F\Delta t$

Elastic Collision

① $m_1V_1 + m_2V_2 = m_1V_1' + m_2V_2'$

② $KE_i = KE_f$

Inelastic Collision

① $m_1V_1 + m_2V_2 = m_1V_1' + m_2V_2'$

② $KE_i \neq KE_f$

perfectly inelastic:

Stick together

$V_1' = \frac{(m_1 - m_2)}{(m_1 + m_2)} \cdot V_1$

$V_2' = \frac{(2m_1)}{(m_1 + m_2)} \cdot V_1$

5 Angular

$x \Rightarrow \theta$
$v \Rightarrow \omega$
$a \Rightarrow \alpha$

Angular Velocity

$\omega = \frac{v}{r}$ $\omega_{avg} = \frac{\Delta\theta}{\Delta t}$

$\omega_{inst} = \frac{d\theta}{dt}$

Rotational kinematics

- $\omega = \omega_0 + \alpha t$
- $\omega_f^2 = \omega_0^2 + 2\alpha\Delta\theta$
- $\Delta\theta = \omega_0 t + \frac{1}{2}\alpha t^2$

Angular acceleration

$\alpha_{avg} = \frac{\Delta\omega}{\Delta t}$

$\alpha_{inst} = \frac{d\omega}{dt}$

Tangential

$V_T = \omega r$

$a_T = \alpha r$

- $T = \frac{2\pi}{\omega}$

- $x = r\cos\theta$
- $y = r\sin\theta$

Direction of angular velocity

RHR: grab onto the pole and curl your fingers in direction of rotation. Thumb equals $\vec{\omega}$

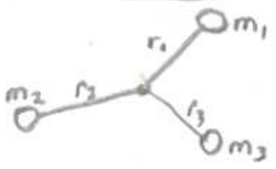
Direction of angular acceleration

points in the direction in the change($\Delta\omega$) in velocity

Rotational Kinetic Energy

$KE_r = \frac{1}{2}I\omega^2$

Moment of Inertia



$I = \Sigma m_i r_i^2$

Point-mass

$I = mR^2$

Torque

$\tau = I\alpha$

Parallel axis thereom

$I = I_{cm} + Md^2$

* if axis is parallel

$I_{sph} = \frac{2}{5}MR^2 + Md^2 = \frac{7}{5}MR^2$

Angular Momentum

$L = I\omega$

if no external torques

$\Sigma\tau = 0$, momentum conserved

$I_i\omega_i = I_f\omega_f$

Counter → (+)

Clockwise → (−)

vertical Spring

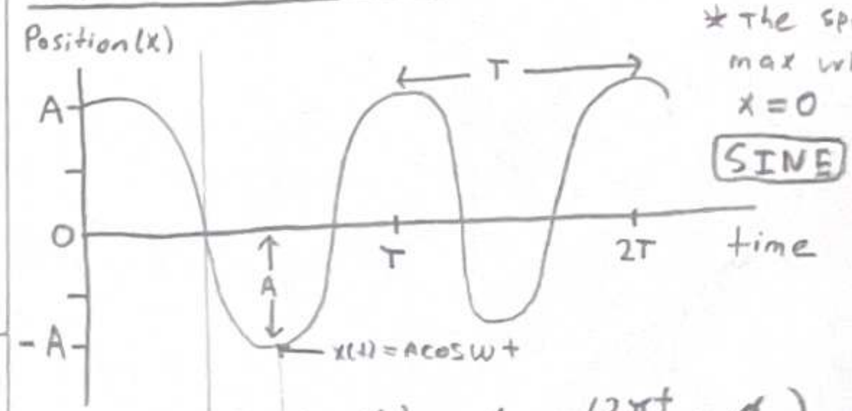
equilibrium position ΔL is

$\Delta L = \frac{mg}{k}$

6 Oscillations / Waves

*The speed is 0 when $x = \pm A$

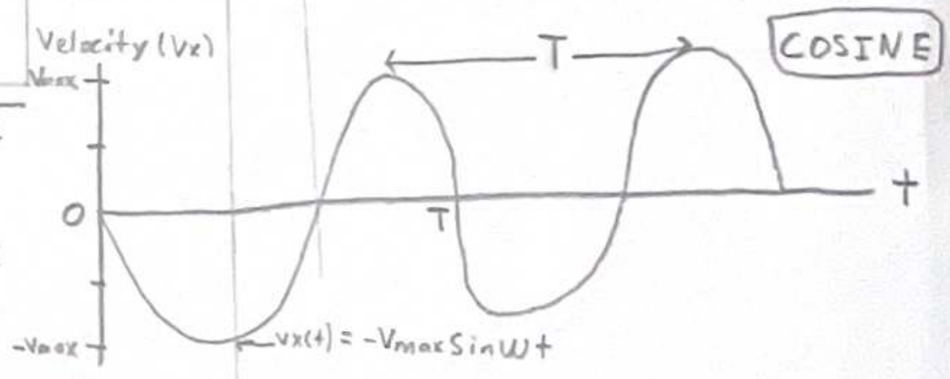
Simple Harmonic motion



*The speed is max when $x = 0$

SINE

$x(t) = A\cos(\omega t + \phi_0) = A\cos\left(\frac{2\pi t}{T} + \phi_0\right)$



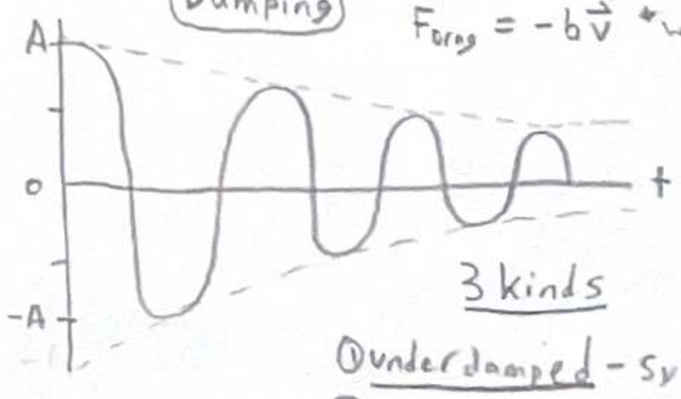
COSINE

$V_x(t) = -V_{MAX}\sin(\omega t + \phi_0)$, $V_{MAX} = \omega A$

$a_x(t) = -\omega^2 x(t) = -\omega^2 A\cos(\omega t + \phi_0)$

Damping

$\vec{F}_{drag} = -b\vec{v}$ *where b is damping constant



3 kinds

① Underdamped – System oscillates

② Critical damping – System reaches equilibrium most quickly

③ Overdamped – hindered by friction from reaching equilibrium

$x(t) = Ae^{-bt/2m}\cos(\omega t + \phi)$

- $\omega = \sqrt{\frac{k}{m}}$
- $f = \frac{\omega}{2\pi}$
- $V_{Max} = \sqrt{\frac{k}{m}} \cdot A$
- $T = 2\pi\sqrt{\frac{m}{k}}$

Waves

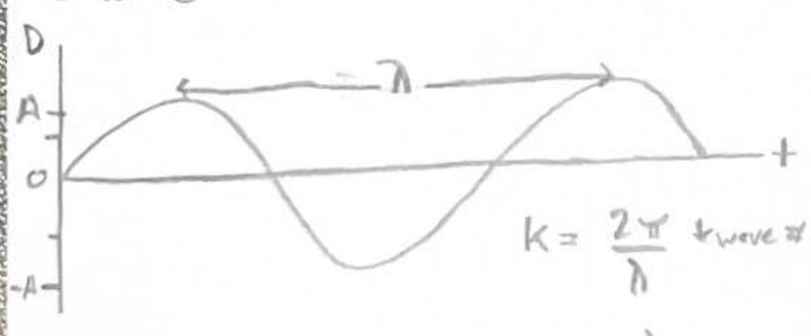
Longitudal ⇒ parallel to wave

Transverse ⇒ Perpendicular to wave

wave speed

$v = \text{distance} / \text{time}$

$= \frac{\lambda}{T} = \lambda \cdot f$

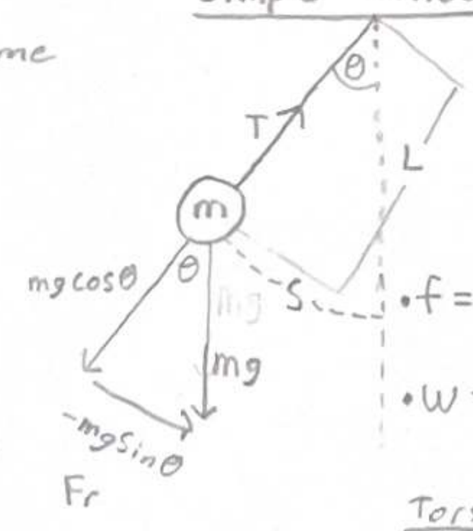


$k = \frac{2\pi}{\lambda}$ ← wave #

$D(x,t) = A\sin(kx - \omega t + \phi_0)$

$D(x,t) = A_0\sin\left(\frac{2\pi}{T} \pm \frac{2\pi x}{\lambda}\right)$

Simple Pendulum $\cdot T = 2\pi\sqrt{\frac{L}{g}}$



*For small angles

$F_r = -mg\theta$

$F_r = -kx$

- $f = \frac{1}{2\pi}\sqrt{\frac{g}{L}}$
- $\omega = \sqrt{\frac{g}{L}}$ $\cdot T = \frac{1}{n}$

Driven Oscillations

$x(t) = A\cos(\omega_d t + \phi)$

Natural frequency

$\omega_0 = \sqrt{\frac{k}{m}}$

$A(\omega) = \frac{F_0}{m\sqrt{(\omega_d^2 - \omega_0^2) + b^2\omega_d^2/m^2}}$

Torsional oscillator

$\omega = \sqrt{\frac{k}{I}}$ *where k equals torsional constant

Resonance

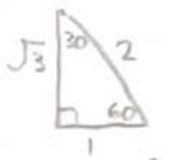
$\frac{1}{2}k(A')^2 = \frac{1}{2}k(A_0)^2 + W_{nc}$

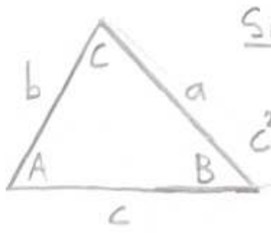
$\frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \frac{1}{2}kA^2 = \frac{1}{2}m(V_{max})^2$

$\cdot k = \frac{f}{\Delta y}$

*Δy = Amplitude

Math





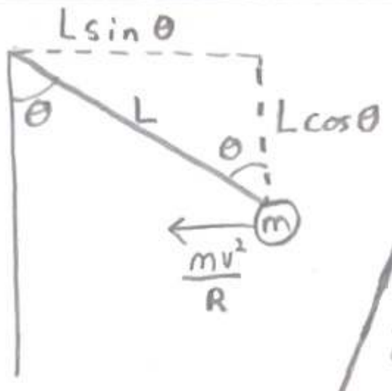
$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$

$c^2 = a^2 + b^2 - 2ab\cos C$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

- $C = 2\pi r$
- $A = \pi r^2$
- $\sin^2 + \cos^2 = 1$
- $\sin(2\theta) = 2\sin\theta\cos\theta$
- π radians = 180°
- $A(\text{sphere}) = 4\pi r^2$
- $V(\text{sphere}) = \frac{4}{3}\pi r^3$

2 A 5.55 kg ball hangs from the top of a vertical pole by a 2.17m long string, the ball is struck causing it to revolve around the pole at a speed of 4.85 m/s in a horizontal circle with the string remaining taut. Calculate the angle, between 0° and 90°, that the string makes with the pole.



SOLUTION:

$T_r = T\sin\theta$, $T_r = \frac{mv^2}{R}$, $R = L\sin\theta$

$T_{ver} = T\cos\theta$, $T\cos\theta = mg$

$\therefore T\sin\theta = \frac{mv^2}{L\sin\theta}$, $T\sin^2\theta = \frac{mv^2}{L}$

$\frac{T\sin^2\theta}{\cos\theta} = \frac{\left(\frac{mv^2}{L}\right)}{mg}$, $\frac{\sin^2\theta}{\cos\theta} = \frac{v^2}{Lg}$

$\frac{1-\cos^2\theta}{\cos\theta} = \frac{v^2}{Lg}$

$Lg\cos^2\theta + v^2\cos\theta - Lg = 0$ QUADRATIC

$\therefore \theta = 53.8°$

2 A 1160 kg racecar is driving around a circular track of radius 149m. In the instant shown, the vehicle has a forward speed of 33.7 m/s and is slowing at a rate of 5.55 m/s². What is the magnitude of the net force acting on the vehicle at this time and the forces direction relative to velocity.

SOLUTION: $F_T = ma_T$, $(1160)(5.55) = 6440N$

$F_c = ma_c = \frac{mv^2}{r}$, $\frac{(1160)(33.7)^2}{149} = 8840N$

$\theta = \tan^{-1}\left(\frac{F_T}{F_c}\right)$

$F_{net} = \sqrt{F_T^2 + F_c^2}$

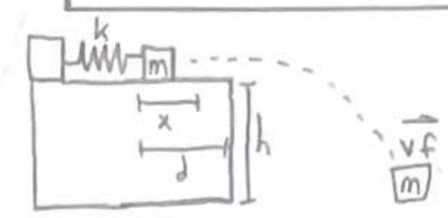
$= \sqrt{(6440)^2 + (8840)^2}$

$= 10900N$

3

Given:
$x = 2m$
$k = 50 N/m$
$d = 3m$
$h = 5m$
$m = 3kg$
$\mu_k = 0.25$

Find:
$V_f = ?$



SOLUTION:

Before: $PE_g + PE_s$ After: $KE + W_{nc}$

$mgh + \frac{1}{2}kx^2 = \frac{1}{2}mv_f^2 + F_f \cdot d$

$(3)(9.81)(5) + \frac{1}{2}(50)(2)^2 = \frac{1}{2}(3)v_f^2 + (3)(9.81)(0.25)(3)$

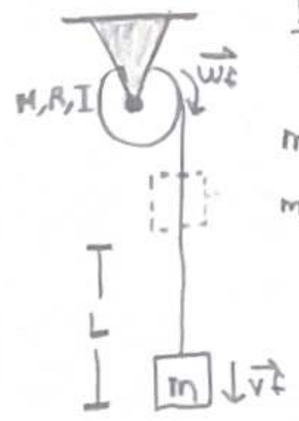
$\therefore \vec{v_f} = 12.25 m/s$

5

Given:
$L = 4m$
$R = 0.5m$
$M = 0.5kg$
$m = 5kg$
$I_{cyl} = \frac{1}{2}MR^2$

Find:
$\vec{v_f} = ?$

*note
$v_{f\,cyl} = v_{f\,block}$



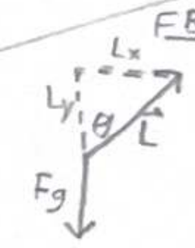
Before: PE_g After: $KE_r + KE_T$

$mgL = \frac{1}{2}I\omega_f^2 + \frac{1}{2}mv_f^2$

$mgL = \frac{1}{2}\left(\frac{1}{2}\right)MR^2\left(\frac{v_f}{R}\right)^2 + \frac{1}{2}mv_f^2$

$\therefore v_f = 8.65 m/s$

FBD:

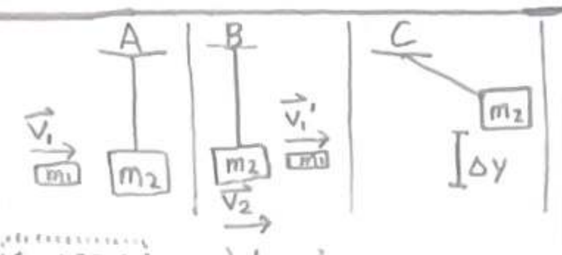


$a_c = \frac{4\pi^2 R}{T}$

4

Given:
$m_1 = 0.007$
$m_2 = 1.5$
$v_1' = 200$
$\Delta y = 0.12$

Find:
$v_1 = ?$



SOLUTION:

Before: KE After: PE

$\frac{1}{2}mv_2'^2 = mg\Delta y$

$\therefore v_2' = 1.53 m/s$

$m_1v_1 + \cancel{m_2v_2} = m_1v_1' + m_2v_2'$

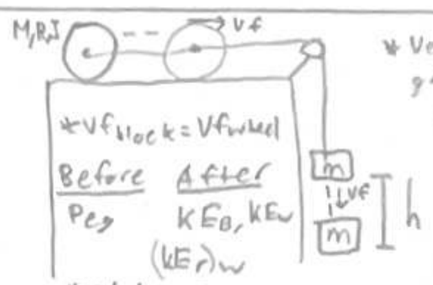
$(0.007)(v_1) = (0.007)(200) + (1.5)(1.53)$

$\therefore v_1 = 528 m/s$

5

Given:
$m = 4$
$M = 8$
$R = 0.5$
$I = 2$
$h = 2$

Find
$v_f = ?$



*$v_{f\,block} = v_{f\,wheel}$

Before: PE_g After: KE_B, KE_W $(KE_r)_W$

*static F_f is conservative

*Vertical Tension greater for pulley with mass

SOLUTION:

$mgh = \frac{1}{2}mv_f^2 + \frac{1}{2}Mv_f^2 + \frac{1}{2}I\omega^2$

$(4)(9.81)(2) = \frac{1}{2}(4)v_f^2 + \frac{1}{2}(8)v_f^2 + \frac{1}{2}(2)\left(\frac{v_f}{0.5}\right)^2$

$\therefore v_f = 2.8 m/s$

2

Given:
$m = 8830kg$
$R = 9.11mi$
$T = 0.121hrs$

Find:
Lift force = ?

$\theta = \tan^{-1}\left(\frac{L_x}{L_y}\right)$

• $\Sigma F_y = L_y + F_g = 0$
• $\Sigma F_x = L_x = ma_c$
• $L_x = \frac{4\pi^2 R}{T}$
• $L_y = mg$

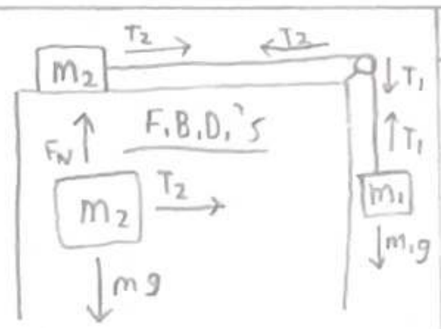
$\vec{L} = \sqrt{L_x^2 + L_y^2} = \sqrt{\left(\frac{4\pi^2 R}{T}\right)^2 + (mg)^2}$

$\therefore \vec{L} = 90.7 kn$

5

Given:
$m_1 = 4$
$m_2 = 3$
$I = 0.5$
$R = 0.3$

Find
$\vec{a} = ?$



$\Sigma F_x = T_2 = m_2 a$

$\Sigma F_y = T_1 - m_1 g = m_1 a$

*note $a_1 = -a_2$

$\alpha = \frac{a}{R^2}$ *For this only

$\Sigma\tau = T_1 - T_2 = I\alpha$

$m_1g - m_1a - m_2a = \frac{Ia}{R^2}$

$(4)(9.81) - (4)(a) - (3)(a) = \frac{(0.5)(a)}{0.3^2}$

$\therefore R = 3.12 m/s$

6 A simple pendulum with a period T = 1 second is attached to the ceiling of an elevator which is initially at rest. Calculate the period of oscillation when the elevator begins to rise upwards with $\vec{a} = 0.6 m/s^2$

SOLUTION: $T = 2\pi\sqrt{\frac{L}{g}}$, Find "L"

$T_{up} = 2\pi\sqrt{\frac{L}{g+0.5}} = 0.975s$

6

Given:
$m = 500kg$
$A = 20cm$
$T = 0.8s$

Find:
@ what position is the blocks speed 1m/s.

Find: what is the spring constant

SOLUTION:

$\frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \frac{1}{2}kA^2$

$X = \sqrt{A^2 - \frac{mv^2}{k}} = \sqrt{A^2 - \left(\frac{v}{\omega}\right)^2}$

$\omega = \frac{2\pi}{0.8} = 7.85$

$x = \sqrt{(0.2)^2 - \left(\frac{1}{7.85}\right)^2} = \pm 15cm$

*block has this speed on either side of equilibrium

SOLUTION: $T = 2\pi\sqrt{\frac{m}{k}}$ *solve for k

6 "Small Angle"

Given:
$m = 300g$
$L = 30cm$
$V = 0.25 m/s$ at lowest point

Find:
Max angle the pendulum reaches

SOLUTION:

$\omega = \sqrt{\frac{g}{L}} = \sqrt{\frac{9.81}{0.30}} = 5.72 rad/s$

at lowest point $V_{MAX} = \omega A$

$A = \frac{V_{MAX}}{\omega} = \frac{0.25}{5.72} = 0.0437M$

$\theta_{max} = \frac{0.0437}{0.30} = 0.146 rad = 8.3°$

Less than 10° so reasonable

6 Swinging Rod Mass not [illegible]

Given:
$L = 0.90m$

Find
Frequency and Period

SOLUTION: $I = \frac{1}{3}ML^2$

$f = \frac{1}{2\pi}\sqrt{\frac{Mgl}{I}}$

$= \frac{1}{2\pi}\sqrt{\frac{Mg(L/2)}{\frac{1}{3}ML^2}}$

$= \frac{1}{2\pi}\sqrt{\frac{3g}{2L}} = 0.64 Hz$, $T = \frac{1}{f}$

$\therefore T = 1.6s$

6 "Damped"

Given:
$m = 500g$
$L = 60cm$
amplitude is observed to decay to half its initial value after 35 oscillations

Find
The time constant "τ"
At what time will the energy have decayed to half its initial value

SOLUTION: After 35 oscillations $X_{MAX} = \frac{1}{2}A$

at t_0, $X_{MAX} = A$

$T = 2\pi\sqrt{\frac{0.60}{9.81}} = 1.55s$

So at 54.25, 35 oscillations have occured

$X_{MAX}(t) = Ae^{-bt/2m} = Ae^{-t/2\tau}$

$\frac{1}{2}A = Ae^{(-54.2)/2\tau}$

$\ln\left(\frac{1}{2}\right) = \frac{-54.2}{2\tau}\ln e$

$\therefore \tau = 39$ seconds

SOLUTION:

*note $b = \frac{m}{\tau} = 0.0513 kg/s$

$E(t) = E_0e^{-t/\tau}$

*E_0 cancels

$\frac{1}{2} = e^{-t_{1/2}/\tau}$

$\ln\left(\frac{1}{2}\right) = \frac{-t_{1/2}}{\tau}\ln e$

$\ln\left(\frac{1}{2}\right) = -\frac{t_{1/2}}{\tau}$

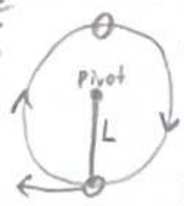
$t_{1/2} = 0.693\tau$

$t_{1/2} = (0.693\tau)(39s) = 27s$ *oscillator loses E faster than A

6 "Driven"

Given
Speed at lowest point is 5m/s
if the pendulum gets pushed at the bottom of the arc

Find
frequency



Before
$y_i = 2L$
$V_i = 0$

After
$y_f = 0$
$V_f = 5$

$\frac{1}{2}v_f^2 = g(2L)$

$= 2gL$

$L = \frac{v_f^2}{2g}$

$= 0.638m$

$f = \frac{1}{2\pi}\sqrt{\frac{g}{L}}$

$= 0.62 Hz$

SOLUTION:

$\frac{1}{2}mv_f^2 + \cancel{mgy_f} = \cancel{\frac{1}{2}mv_i^2} + mgy_i$

6 "Wave"

A sinusoidal wave with amplitude 1cm and frequency 100Hz travels at 200m/s in the positive x direction. At t = 0, the point x = 1m is on a crest of the wave

Find
• A
• v
• λ
• k
• f
• ω
• T
• φ
and write displacement equation

What we know from Above
• A = 1cm
• v = 200m/s
• f = 100 Hz

SOLUTION:

$\lambda = \frac{v}{f} = 2m$

$k = \frac{2\pi}{\lambda} = 3.14 rad/m$

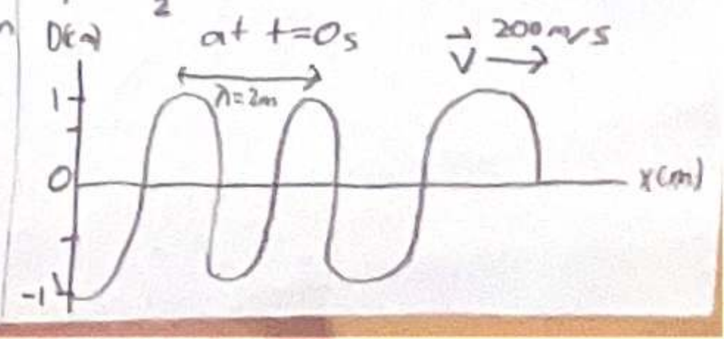
$\omega = 2\pi f = 628 rad/s$

$T = \frac{1}{f} = 0.01s$

$D(x_0, t_0) = A = A\sin[k(1m) + \phi_0]$

$k(1) + \phi = \frac{\pi}{2}$

$\therefore \phi = \frac{-\pi}{2}$

Wave Equation

$D(x,t) = (1cm)\sin[(3.14)x - (628)t - \frac{\pi}{2}]$